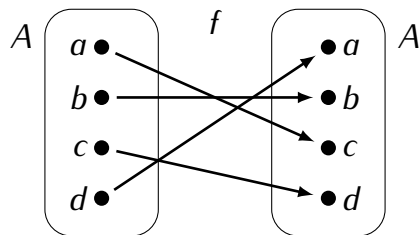
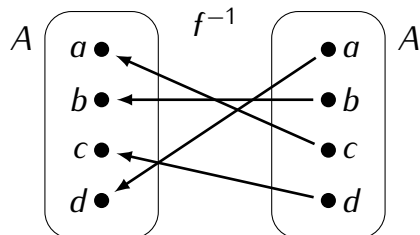


Definition 10.1

A *permutation* of a set A is a function $f: A \rightarrow A$ which is a bijection.



Note. For every permutation f we have the inverse function f^{-1} such that $f \circ f^{-1}(x) = x$ and $f^{-1} \circ f(x) = x$ for all $x \in A$.

**Definition 10.2**

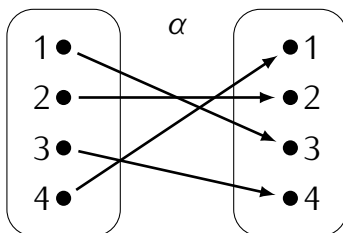
Let A be a set. The *permutation group* of A is a group $S(A)$ defined as follows:

- **Elements of $S(A)$:** permutations $f: A \rightarrow A$.
- **Group operation:** composition of functions $g \circ f$.
- **The identity element:** the function $\varepsilon: A \rightarrow A$, $\varepsilon(x) = x$ for all $x \in A$.
- **The inverse of f :** the inverse permutation f^{-1} .

Definition 10.3

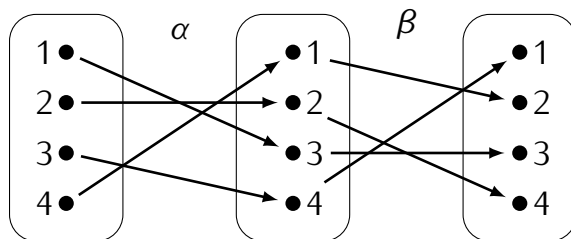
For $n \geq 1$ the group S_n is the group of permutations of the set $A = \{1, 2, \dots, n\}$. This group is called the *symmetric group on n letters*.

Matrix notation of permutations:



$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{bmatrix}$$

Composition:



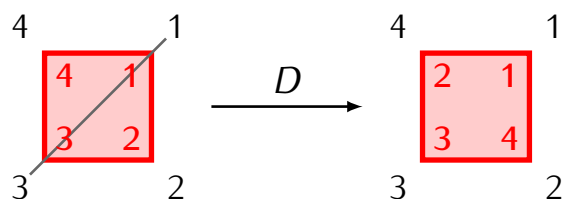
$$\begin{aligned} \beta \circ \alpha &= \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{bmatrix} \circ \begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{bmatrix} \end{aligned}$$

Theorem 10.4

For any $n \geq 1$ we have $|S_n| = n!$

Dihedral groups and permutation groups

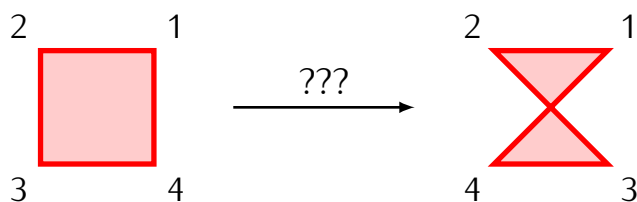
Let P_n be a regular polygon with n vertices. Label the vertices with numbers $1, 2, \dots, n$. Since every symmetry of P_n sends vertices to vertices, it defines a certain permutation of vertices:



$$D = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 3 & 2 \end{bmatrix}$$

Since composition of symmetries corresponds to composition of permutations of vertices, we can identify the dihedral group D_n with a subgroup of the group of permutations S_n . Note that not every permutation in S_n comes from a symmetry of P_n . E.g.:

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{bmatrix}$$



Note. The groups S_n are non-abelian for $n > 2$, e.g:

$$\begin{aligned} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}^{\alpha} \circ \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix}^{\beta} &= \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}^{\alpha \circ \beta} \\ \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix}^{\beta} \circ \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}^{\alpha} &= \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix}^{\beta \circ \alpha} \end{aligned}$$